

DiffCMR

Fast Cardiac MRI Reconstruction with Diffusion Probabilistic Models

Paper

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Institution

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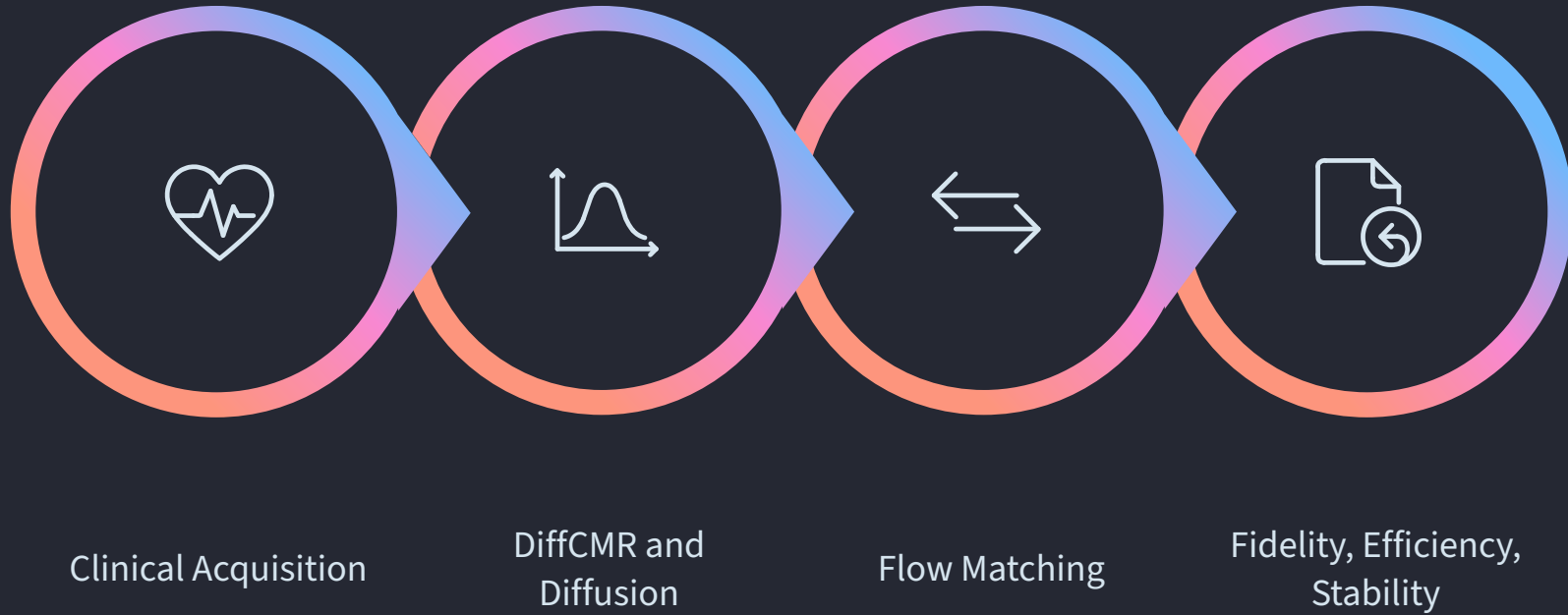
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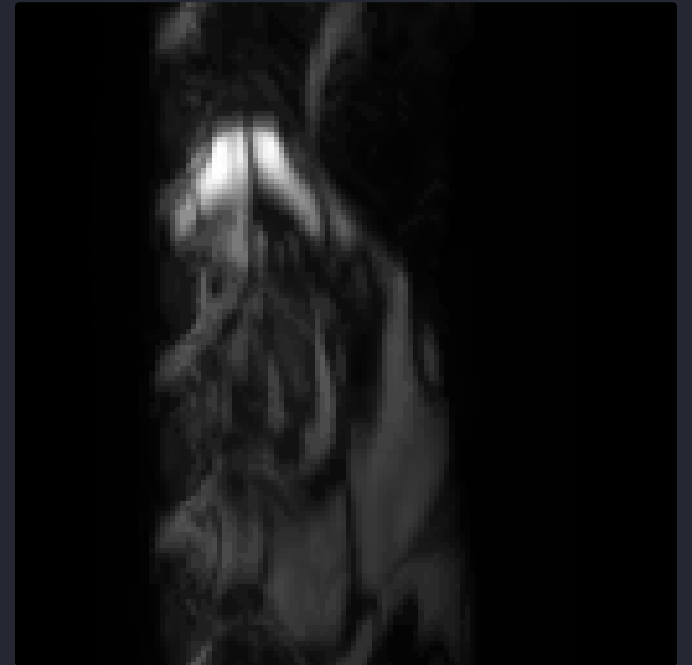
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Introduction: Overall



Background: Clinical context

- **The Clinical Gold Standard**
- A full exam typically takes **30 to 45 minutes** (excluding clinical delays!)
- Relying on **breath-holds** to maintain the required breath-hold consistency is difficult for patients suffering from **advanced cardiac diseases or respiratory distress**.
- **Throughput Bottleneck.**



Background: The Rise of Diffusion Models in MRI Reconstruction

- **From Regression to Generation**
 - **Diffusion Models (DDPM)** have revolutionized the field by learning to *generate* data from noise, preserving high-frequency details and fine textures.
- **Adaptation to MRI: Conditional Diffusion**
 - In our task, the model isn't generating a random heart, it is **conditioned** by the under-sampled data.

¹Ho, J., Jain, A., & Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. *arXiv preprint arXiv:2006.11239*.

The DiffCMR Paradigm

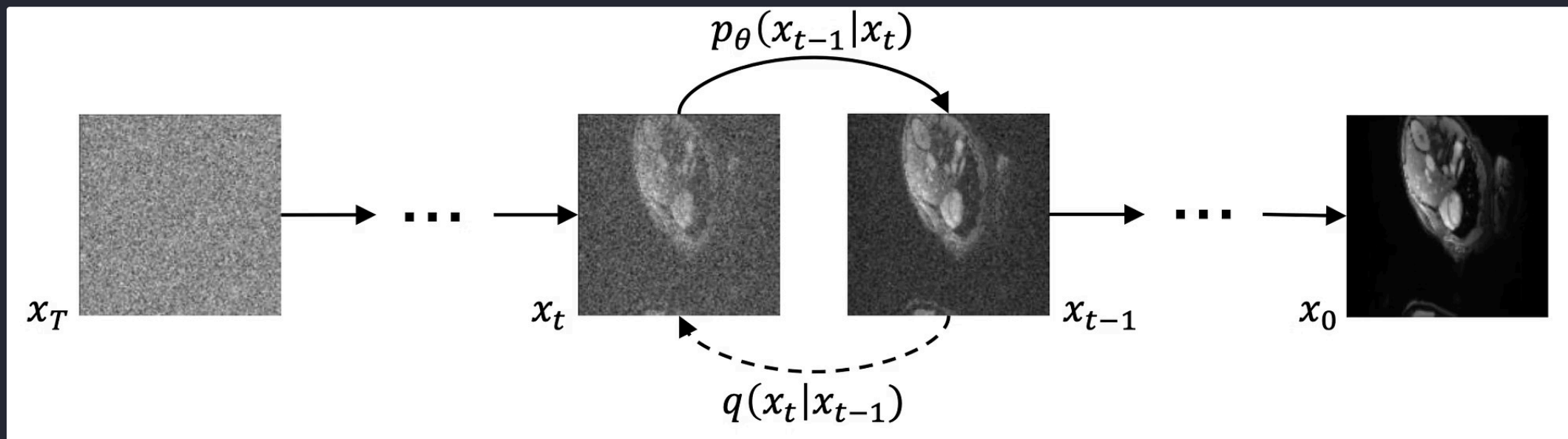


Fig 1: Forward and backward denoising diffusion processes

The DiffCMR Paradigm

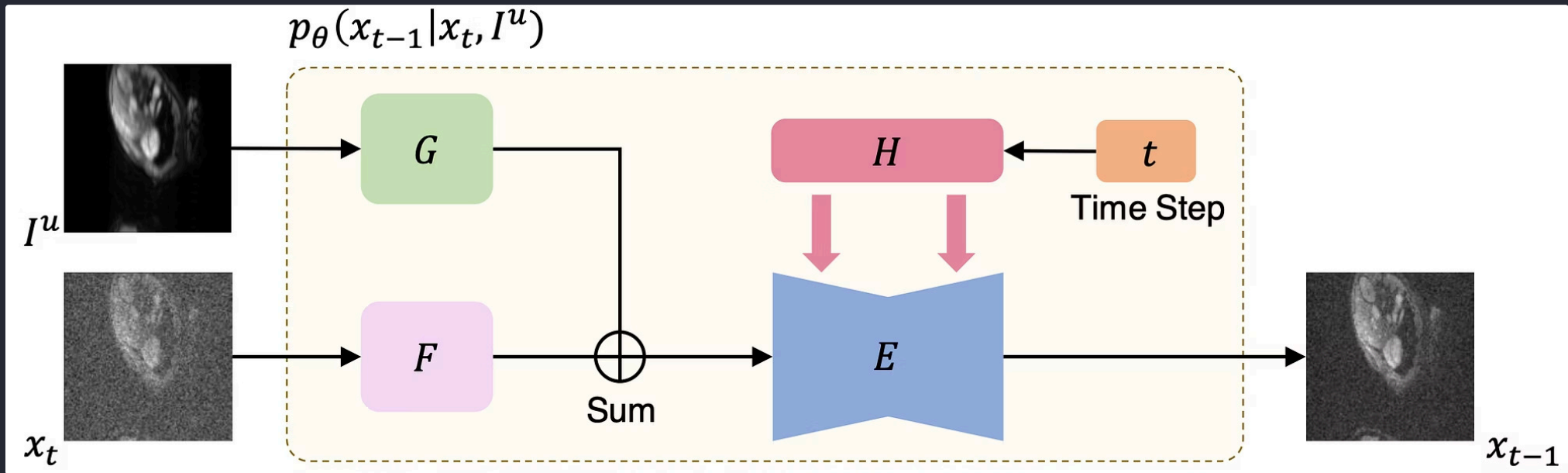
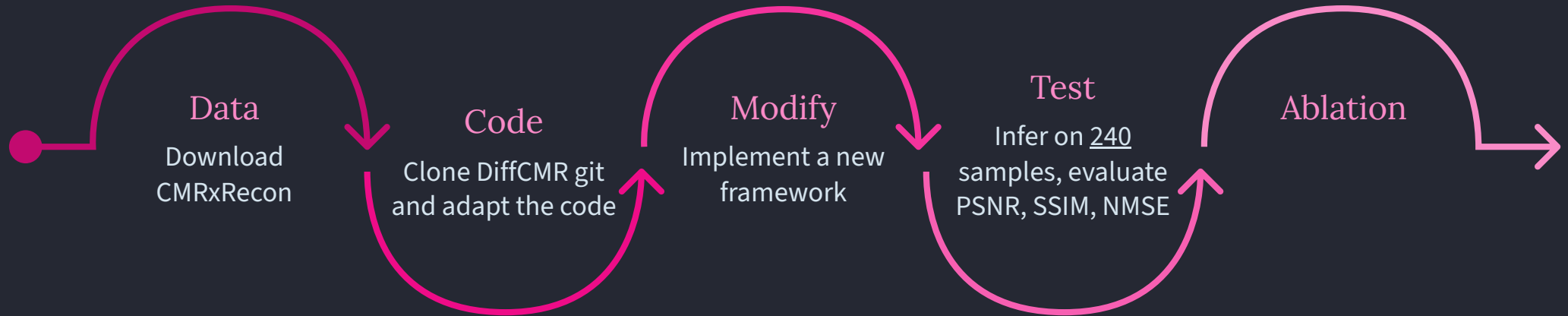


Fig. 2: The figure above illustrates the pipeline of our proposed DiffCMR. F and G encode the noisy signal x_t and the under-sampled image I^u , respectively. H encodes timestamp t to obtain timestamp embeddings. E is a modified U-net that receives both summed features and timestamp embeddings for denoising.

How We Reproduced the Results



Process & Engineering: Reproduction

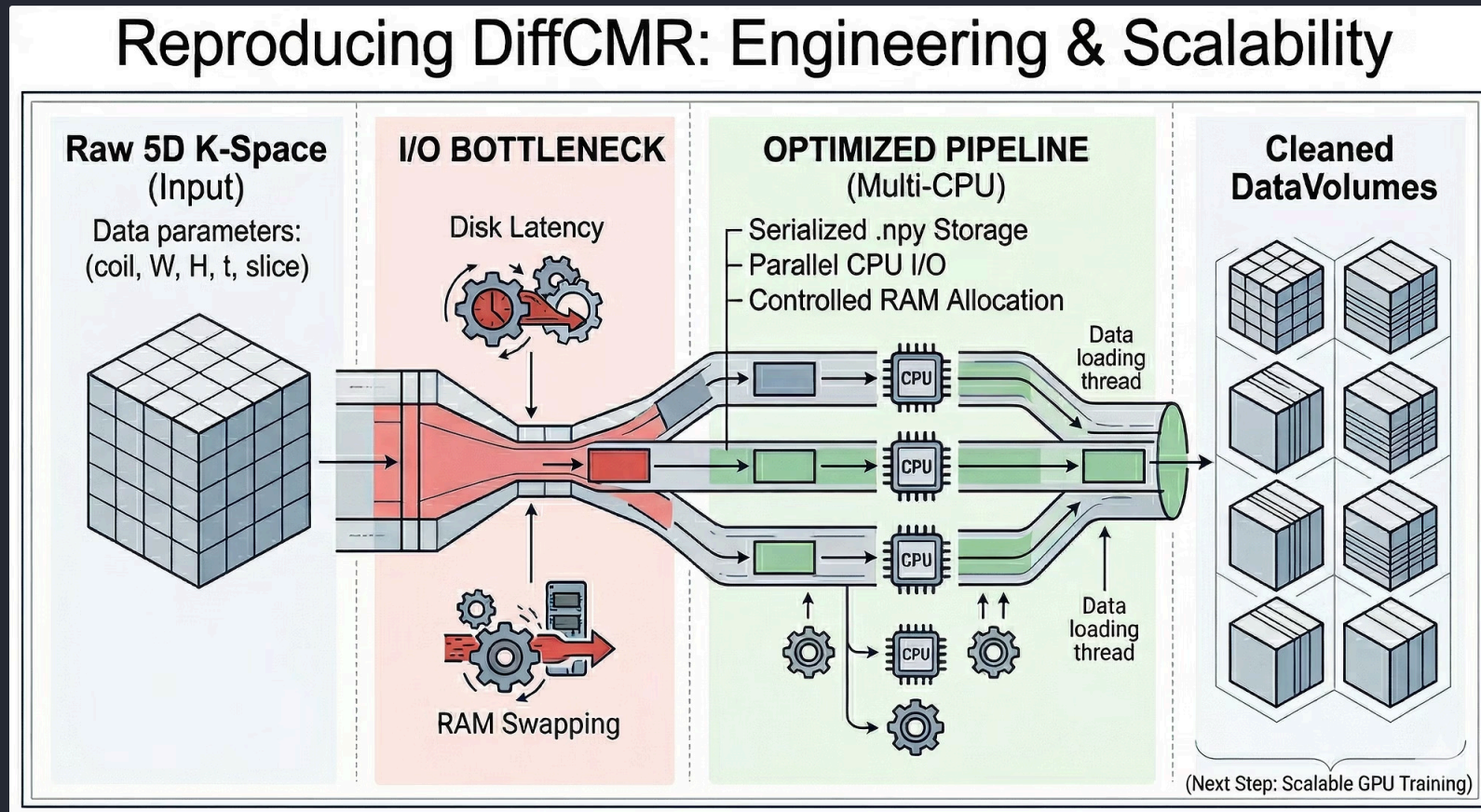


Fig 3: Rewritten data pipeline

The Flow Matching Hypothesis

Our Hypothesis :

- The model should **require fewer steps**.
- The generated results should be **more deterministic and more robust**.

Code Modifications :

- Developed an entirely new, Python module for Flow Matching.
- Implemented a custom **Euler ODE Solver** from scratch to compute the deterministic trajectories during the inference phase.
- Created strictly independent training, and hyperparameter (`argparse`) scripts to guarantee a perfectly fair and mathematically sound comparison against DiffCMR.

Results: Qualitative

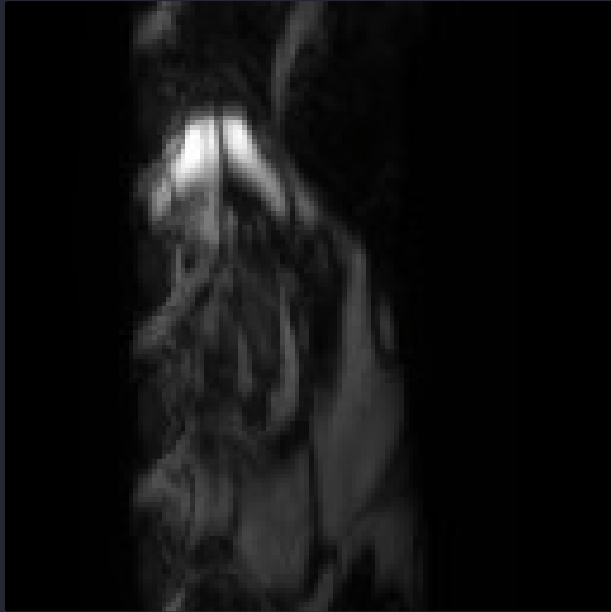


Fig 4.A: Input aliased image



Fig 4.B: Ground truth

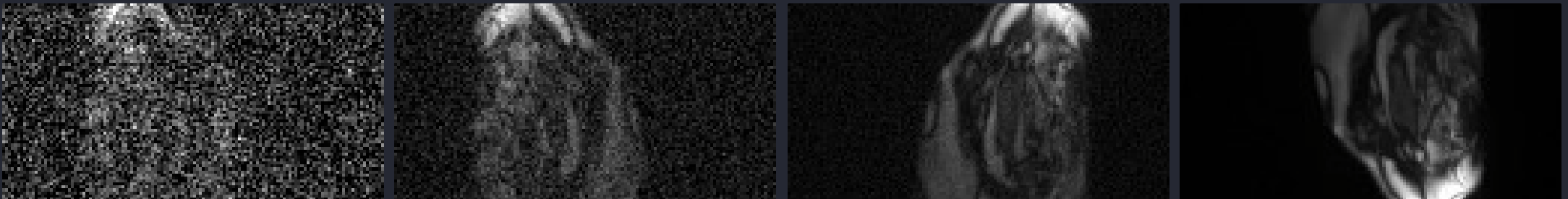


Fig 4.C: Evolution at time step 10k, 20k, 30k, 150k

Results: Qualitative

Table 1: Results on Task 1-AccFactor04: Flow Matching drastically reduces inference time while preserving high quantitative fidelity.

#Steps T	Batch Size	Time/img	PSNR $\uparrow$	SSIM $\uparrow$	NMSE $\downarrow$
Raw	8	-	28.79	0.8150	0.2187
Diffusion (Their DiffCMR Original)					
1000	8	-	36.10	0.9277	0.0346
Diffusion (Our DiffCMR Reproduction)					
1000	8	105.08 s	37.13	0.9594	0.0096
Flow Matching (Our Novel Extension)					
40	8	4.332 s	36.77	0.9556	0.0105

Performance Shift

Time/img: 105s $\rightarrow$ 4.3s

-96% Time

PSNR: 37.13 $\rightarrow$ 36.77
Negligible PSNR Drop (-0.97%)

Fig 5: Global results

Results: Qualitative

#Steps T	Batch Size	Time/img	PSNR $\uparrow$	SSIM $\uparrow$	NMSE $\downarrow$
Raw	8	-	28.79	0.8150	0.2187
Diffusion (DiffCMR)					
20	8	-	25.52	0.3664	0.3830
100	8	-	29.86	0.6006	0.1393
500	8	-	35.39	0.9010	0.0399
1000	8	-	36.10	0.9277	0.0346
1000	8	105.0879	37.128	0.9594	0.0096
Flow Matching (Ours)					
1	16	0.1094	36.477	0.9138	0.0111
4	16	0.4209	36.829	0.9560	0.0103
10	16	1.1008	36.824	0.9559	0.0103
40	8	4.3315	36.770	0.9556	0.0105
100	16	10.4486	36.545	0.9442	0.0109

T-Dependence Analysis

Flow Matching: High Stability

PSNR stays extremely stable:

36.477 (T=1) $\rightarrow$ 36.829 (T=4) $\rightarrow$ 36.545 (T=100)

Robust performance at any step count.

Fig 6: Ablation study on steps

Results: Qualitative

#Rounds R	Batch Size	Time/img	PSNR $\uparrow$	SSIM $\uparrow$	NMSE $\downarrow$
Raw	8	-	28.79	0.8150	0.2187
Diffusion (DiffCMR)					
1	8	-	33.89	0.8491	0.0561
2	8	-	35.25	0.8994	0.0417
4	8	-	36.10	0.9277	0.0346
8	8	-	36.68	0.9430	0.0305
4	8	105.0879	37.128	0.9594	0.0096
Flow Matching (Ours)					
1	8	1.1079	36.356	0.9463	0.0115
4	8	4.3315	36.770	0.9556	0.0105
8	16	8.4371	36.833	0.9570	0.0103

PSNR Stability Analysis

Diffusion R-Dependence:

R=1: PSNR 33.89
R=8: PSNR 36.68 **+2.8 dB Jump!**

Highly unstable at low R.

Fig 7: Ablation study on Rounds

Conclusion: Flow Matching Emerges as the Superior Paradigm

Stability

Less stochasticity, leading to more consistent generation.

~1000

DDPM Steps

PSNR ~37.13 dB

Scalability

Requires far fewer inference steps.

~40

Flow Matching Steps

PSNR ~36.77 dB

Accuracy

Delivers comparable metrics.

50–200x

Faster Inference

Major speedup with similar or improved quality

The future of generative models in medical imaging will favor paradigms that combine high fidelity with clinical efficiency — and Flow Matching is a strong step in that direction.

Thank you for your attention!

Do you have any questions?

Forward Process: Adding Noise Step by Step

The Closed-Form Trick

A key property of the Gaussian forward process is that we can sample **any noisy step directly** without simulating all intermediate steps:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$. This makes training extremely efficient — no sequential simulation needed.

Variance Schedule β_t

Ho et al. use a **linear schedule** from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$, with **T = 1000** timesteps. As $t \rightarrow T$, the signal-to-noise ratio approaches zero and $x_T \approx \mathcal{N}(0, I)$.

Connection to Score Matching

DDPM is equivalent to **denoising score matching** with Langevin dynamics. The noise predictor ε_θ implicitly estimates the gradient of the log-density (score function) of the noisy data distribution at each noise level.

01

Clean Image x_0

Original data sample from the training distribution — e.g., a fully-sampled MRI slice.

03

Pure Noise x_T

After $T=1000$ steps, the image is entirely Gaussian noise — the starting point for generation.

02

Noisy x_t

Intermediate noisy version. The closed-form trick lets us jump directly here from x_0 .

04

Reverse Denoising

The U-Net predicts ε at each step, iteratively reconstructing the clean image from noise.

Reverse Process: Learning to Denoise

The Reverse Step

Given a noisy sample x_t , the model predicts the previous step x_{t-1} via a learned Gaussian transition:

$$p_{\theta}(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \sigma_t^2 I)$$

The mean μ_{θ} is derived from the predicted noise ϵ_{θ} , and σ_t is fixed (not learned in the original paper). This formulation connects to **denoising score matching** + **Langevin dynamics** sampling.

Key Results (Ho et al. 2020)

3.17

FID Score

State-of-the-art on CIFAR-10 at the time of publication.

9.46

Inception Score

Unconditional CIFAR-10 generation quality.

1000

Timesteps T

Number of denoising steps — the main bottleneck for inference speed.

📌 ⚠️ **The Speed Problem:** DDPM requires $T = 1000$ sequential neural network evaluations to generate a single sample. For medical imaging applications, this is the fundamental bottleneck that motivates both DiffCMR's design choices and the later move to flow matching.

Architecture & Technical Design

Data Pipeline

- **iFFT transform:** inverse Fast Fourier Transform converts raw k-space data (fully-sampled & under-sampled) into 2D image slices.
- **Zero-padding:** k-space zero-padded to 512 before iFFT, slices resized to 128×128 to accelerate training.
- **Conditioning:** the under-sampled image slice is concatenated to the noisy input — the U-Net receives both to predict the added noise.
- **Multi-round ensembling:** during inference, multiple DDPM sampling runs are averaged to reduce stochastic variance and stabilize output quality.

Tasks & Dataset

Validated on the **MICCAI 2023 CMRxRecon** dataset:

- **Task 1 — Cine Reconstruction:** dynamic cardiac motion imaging
- **Task 2 — T1/T2 Mapping:** quantitative tissue characterization
- **120 training cases**, 60 validation cases per task
- **Coil types:** single-coil and multi-coil variants
- **AccFactors:** ×4, ×8, ×10

Evaluation Metrics

PSNR (Peak Signal-to-Noise Ratio), **SSIM** (Structural Similarity Index), and **NMSE** (Normalized Mean Square Error). Compared against U-Net and cGAN baselines.